INCH–POUND

MIL–DTL–51227C
25 September 2013

SUPERSEDING
MIL–P–51227B
27 June 1972

DETAIL SPECIFICATION

PUMP, CENTRIFUGAL, DECONTAMINATING APPARATUS

Inactive for new design after 14 March 1996

This specification is approved for use by the U.S. Army Edgewood Chemical Biological Center and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a pump for a decontaminating apparatus.

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 or 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 or 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

Comments, suggestions, or questions on this document should be addressed to: U.S. Army Edgewood Chemical Biological Center, ATTN: RDCB–DEA–S, 5183 Blackhawk Road, Aberdeen Proving Ground, MD 21010–5424 or emailed to APGR–EcbcSpecsTeam@conus.army.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil.

AMSC N/A FSC 4230

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DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL−DTL−12468 − Decontaminating Agent, STB
MIL−DTL−51027 − Antisetting Compound, Decontaminating Slurry, M2

DEPARTMENT OF DEFENSE STANDARDS

MIL−STD−1916 − Preferred Methods for Acceptance of Product

(Copies of these documents are available online at http://quicksearch.dla.mil/ or from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111−5094.)

2.2.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

U.S. ARMY EDGEWOOD CHEMICAL BIOLOGICAL CENTER

DRAWINGS

5−45−2835 − Pump Assembly

(Copies of these documents are available from U.S. Army Edgewood Chemical Biological Center, ATTN: RDCB−DEA−D, 5183 Blackhawk Road, Aberdeen Proving Ground, MD 21010−5424.)

2.3 Non−Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

AMERICAN INSTITUTE OF CHEMICAL ENGINEERS

AICHE E−31 − Centrifugal Pumps (Newtonian Liquids): A Guide to Performance Evaluation, third edition

(Copies of this document are available from http://www.aiche.org/ or American Institute of Chemical Engineers (AIChE), 3 Park Avenue 19 Fl, New York, NY 10016−5991.)

2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein (except for related specification sheets), the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

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3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection in accordance with 4.2.

3.2 Design and construction.

3.2.1 Materials. All materials cited on Drawing 5−45−2835 and its subsidiary drawings shall conform to the specifications listed thereon or to the specific characteristics set forth on the drawings. Material not definitely specified shall be of a quality regularly used by manufacturers of pumping equipment.

3.2.2 Components. All components of the item shall conform to the specifications and drawings listed on Drawing 5−45−2835 and subsidiary drawings.

3.2.3 Manufacture and assembly. The item shall be manufactured and assembled as shown on Drawing 5−45−2835.

3.3 Functioning. The pump shall meet the following requirements when running at 4400 ± 100 revolutions per minute (rpm) for a minimum period of 15 minutes when tested as specified in 4.4.1.

(a) Discharge 100 gallons per minute (gpm) minimum of water at a discharge head of 205 feet ± 1 foot with a minimum static lift of 15 feet.
(b) Operate smoothly and not leak.
(c) Prime in a maximum of 4 minutes by using no more than 9 gallons of water.

3.4 Performance. The pump running at 4400 ± 100 rpm shall be capable of delivering a minimum of 50 gpm of STB slurry, made in accordance with 4.4.2, at a discharge head of 290 feet ± 1 foot and a minimum dynamic lift of 10 feet. The pump shall also deliver a minimum of 100 gpm of water at a discharge head of 205 feet ± 1 foot with a minimum static lift of 15 feet when running at 4400 ± 100 rpm. The total head at zero flow shall be a minimum of 300 feet. The required horsepower (hp) to drive the pump shall not exceed 20 hp when running at 4400 ± 100 rpm and pumping a minimum of 50 gpm of STB slurry or an equivalent mass flow of water at a minimum of 140 pounds per square inch (psi) at the discharge gage.

3.5 Endurance. The pump shall be operated for a minimum of 250 hours with only minor adjustments. The pump shall operate smoothly without overloading or leakage when tested as specified in 4.4.3.

3.6 Performance curve. The discharge head versus capacity curve of the pump when tested as specified in 4.4.4 shall slope continuously from high head downward toward high capacity. The curve may approach constant head as the capacity increases but the head may never in-

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crease. The head capacity coordinates indicated in 3.3 shall be on or below the pump performance curve.

3.7 Weight. The assembled pump weight shall not exceed 100 pounds when tested as specified in 4.4.5.

3.8 Cavitation. Cavitation shall not occur when tested as specified in 4.4.6.

3.9 First article. When specified (see 6.2), a sample shall be subjected to first article inspection in accordance with 4.2.

3.10 Workmanship. The pump shall be free from contamination (foreign matter, oil or grease) and damage such as cracks , burrs and sharp edges.

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

(a) First article inspection (see 4.2)
(b) Conformance inspection (see 4.3)

4.2 First article inspection.

4.2.1 Sample. The first article sample shall consist of two pumps manufactured using the same methods, materials, equipment, and processes as will be used during regular production. The first article sample shall be submitted for inspection and approval in accordance with the terms of the contract.

4.2.2 Inspection procedure. The first article sample items shall be subjected to the inspection procedures specified in table I.

TABLE I. First article inspection procedures.

Inspection procedure	Number of sample units	Requirement paragraph
Performance	2	3.4
Endurance	2	3.5
Performance curve	2	3.6
Weight	2	3.7
Cavitation[1]	2	3.8

[1]Destructive test.

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4.2.3 Acceptance criteria. If any first article sample item fails to comply with any of the applicable requirements, the first article sample shall be rejected. The Government reserves the right to terminate inspection upon any failure to comply with any of the requirements. The contractor shall obtain written approval from the contracting activity prior to proceeding with regular production.

4.3 Conformance inspection.

4.3.1 Lotting. A lot shall consist of the items produced by one manufacturer, at one plant, from the same materials, under the same manufacturing conditions, and at the same time.

4.3.2 Sampling. Sampling shall be conducted in accordance with the classification of characteristics in 4.3.5. Samples shall be selected at random.

4.3.3 Inspection procedure. Sample items shall be inspected in accordance with the classification of characteristics in 4.3.5. Failure of any sample item to conform to any characteristic in the classification of characteristics based on the sampling and acceptance criteria specified therein shall be cause for rejection of the lot represented.

4.3.4 Critical characteristics. Unless otherwise specified in the classification of characteristics, the sampling and acceptance criteria for critical characteristics shall be in accordance with the **Critical characteristics** paragraph of MIL−STD−1916.

4.3.5 Classification of characteristics. Conformance inspections shall be as specified in the following classification of characteristics paragraphs. When verification levels (VLs) are specified for selection of sample sizes and acceptance criteria, the attributes sampling plan of MIL−STD−1916 shall apply unless otherwise specified.

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CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE		DRAWING NUMBER
4.3.5(a)	Pump assembly	SHEET 1 OF 1	5–45–2835

CATEGORY	CHARACTERISTIC	SAMPLING AND ACCEPTANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD
Critical	Functioning (less prime test)	100% inspection; acc 0, rej 1	3.3	4.4.1
Major				
101	Grease fitting present	VL–III	3.2	VI
102	Breather unobstructed	VL–III	3.2	VI
103	Prime time test	VL–III	3.3	4.4.1
104	Workmanship	VL–III	3.10	VI

NOTES:

CE – Commercial inspection equipment
VI – Visual inspection

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CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE		DRAWING NUMBER
4.3.5(b)	Tank pump	SHEET 1 OF 1	5−45−2756

CATEGORY	CHARACTERISTIC	SAMPLING AND ACCEPTANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD
Critical	None defined			
Major				
101	Mounting holes diameter	VL−III	3.2	CE
102	Mounting holes location	VL−III	3.2	VI
103	Thread (2−11−1/2NPT)	VL−III	3.2	CE
104	Thread (1/2−14NPT)	VL−III	3.2	CE
105	Marking correct and evident	VL−III	3.2	CE

NOTES:

CE − Commercial inspection equipment

VI − Visual inspection

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CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE		DRAWING NUMBER
4.3.5(c)	Cover, pump tank Shaft	SHEET 1 OF 1	5−45−2776 5−45−2751

CATEGORY	CHARACTERISTIC	SAMPLING AND ACCEPTANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD
Critical	None defined			
Major				
101	Thread (3/8−18 NPT) (5−45−2776)	VL−III	3.2	CE
102	Shaft diameter (0.875) (5−45−2751)	VL−III	3.2	CE
103	Keyway depth (0.245) (5−45−2751)	VL−III	3.2	CE
104	Width (0.2487) (5−45−2751)	VL−III	3.2	CE
105	Radius (0.887) (5−45−2751)	VL−III	3.2	CE

NOTES:

CE − Commercial inspection equipment

VI − Visual inspection

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4.4 Tests.

4.4.1 Functioning. The pump shall be operated at the required speed for the specified time pumping water. Tests shall be conducted in accordance with procedures outlined in ***section 500.0 of the American Institute of Chemical Engineers Centrifugal Pump Manual***. The pump shall be observed for smoothness of operation and leakage. Discharge capacity, discharge head, static lift and time required to prime the pump shall be recorded or calculated as outlined in ***section 600.0 of the American Institute of Chemical Engineers Centrifugal Pump manual.***

4.4.2 Performance. The performance of the pump shall be as specified when connected to a suitable drive and supply tank. The horsepower shall be determined when pumping slurry and tested in accordance with ***section 500.0 or calculated in accordance with section 600.0 of the American Institute of Chemical Engineers Centrifugal Pump Manual.*** The slurry, when required for the above tests, shall have the following proportions:

(a) Two hundred and twenty−five gallons of water.
(b) Twelve and one half pounds of decontaminating slurry antiset M2 conforming to MIL−DTL−51027 (antiset shall be added to the water before STB).
(c) Thirteen hundred pounds of decontaminating agent STB conforming to MIL−DTL−12468.
(d) 144 ounces of Antifoam (Antifoam Xiameter AFE 0200) or purchased from the Government Supply system using NSN: 6850−00−950−6489.

4.4.3 Endurance. After completion of the test in 4.4.2 the pump shall be operated pumping slurry from and to a tank. Fresh slurry mixture shall be prepared whenever required, but at least every 16 hours. The pump shall be flushed and cleaned whenever shutdown for more than 1 hour. The pump shall run at 4400 ± 100 rpm and discharge the slurry through an orifice 0.562 ± 0 .010 inch in diameter. Necessary precautions shall be taken to keep the temperature of the mixture being pumped between 85˚F and 135˚F while testing. The slurry shall have the following proportions for the above test:

(a) Two hundred and twenty−five gallons of water.
(b) 72 ounces of Antifoam (Antifoam Xiameter AFE 0200) or purchased from the Government Supply system using NSN: 6850−00−950−6489.
(c) Six and one quarter pounds of decontaminating slurry antiset M2 conforming to MIL−DTL−51027 (the antiset shall be added to the water before the STB).
(d) Six hundred and fifty pounds of decontaminating agent STB conforming to MIL−DTL−12468.

4.4.4 Performance curve. Performance of the pump shall be determined using plain water. Test shall be conducted in accordance with procedures outlined in ***section 500.0 of the American Institute of Chemical Engineers Centrifugal Pump Manual;*** graphical presentation of the pump performance data shall be in accordance with ***section 700.0 of the American Institute of Chemical***

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Engineers Centrifugal Pump Manual. A minimum of six points including no flow and maximum flow conditions shall be obtained for the discharge head−capacity curve.

4.4.5 Weight. The weight of the pump shall be as specified when weighed on a suitable scale accurate to within 1 pound.

4.4.6 Cavitation. The impeller shall be visually examined for cavitation after testing as specified in 4.4.3. If necessary for examination, the impeller shall be cut perpendicular to the axis of rotation. If evidence of cavitation is present, the design is unsatisfactory and shall be rejected. If the impeller does not show evidence of cavitation and the pump meets the requirements of 3.8, the pump may be rebuilt to its new condition.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When actual packaging of materiel is to be performed by DoD or in−house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain requisite packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD−ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The pump covered by this specification is intended for use in the M12A1 decontaminating apparatus. The pump may also be used in other applications to pump various fluids.

6.2 Acquisition requirements. Acquisition documents should specify the following:

(a) Title, number, and date of this specification.
(b) First article:

(1) Time allowed for contractor submission of samples for Government test and evaluation after award of contract when testing is performed by the Government.
(2) Name and address of test facility and shipping instructions when testing is performed by the Government.
(3) Time required for the Government to notify the contractor whether or not to proceed with production.

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(c) Packaging requirements (see 5.1).

6.3 Subject term (key word) listing.

STB – Super Tropical Bleach

6.4 Changes from previous issues. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

Custodians:
Army – EA
DLA – GS

Preparing activity:
Army – EA
(Project 4230−2013−001)

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